

CHEMISTRY | UNIT 02 | LESSON 02

Protons, Neutrons, Electrons, and Atomic Structure

Checkpoint Submission

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.02	Submission:	Checkpoint Submission

Directions

Complete this sheet independently after finishing the lesson review. Show every calculation. Upload this PDF to U2 L2 Checkpoint Submission.

Reference

Particle	Charge	Relative mass	Location
proton	+1	about 1 amu	nucleus
neutron	0	about 1 amu	nucleus
electron	-1	very small	electron cloud

Part A - Determine particle counts

Atom/ion	Atomic #	Mass #	Charge	p+	n0	e-
Magnesium-24	12	24	0			
Aluminum-27 ion	13	27	+3			
Fluorine-19 ion	9	19	-1			

Show the neutron and electron calculations for all three rows.

Part B - Atomic model

Draw a labeled model of the aluminum ion above. Show the nucleus, 13 protons, 14 neutrons, and the correct number of electrons. Your drawing may use grouped symbols rather than individual circles.

DRAW MODEL HERE

Part C - CER explanation

Claim: Changing electrons changes charge but does not change element identity. Use one row from Part A as evidence and explain why the claim is correct.
